Exam 1 Review: Scalar Flows and Bifurcations

Tuesday, October 6, 2026

Idea: Scalar-flow analysis

For a parameter-dependent scalar equation

$$\dot{x} = f(x; r),$$

use the chain

$$\boxed{\text{state space}} \longrightarrow \boxed{f(x;r)=0} \longrightarrow \boxed{\text{sign } f} \longrightarrow \boxed{\text{stability and limits}} \longrightarrow \boxed{\text{parameter dependence}}.$$

Recall: Hyperbolic equilibrium test

Let x^* be an equilibrium of $\dot{x} = f(x; r)$, where f is continuously differentiable with respect to x. If $f_x(x^*; r) \neq 0$, then x^* is **hyperbolic**, and

$$f_x(x^*; r) < 0 \Rightarrow x^* \text{ is asymptotically stable}, \qquad f_x(x^*; r) > 0 \Rightarrow x^* \text{ is unstable}.$$

Observation: Nonhyperbolic scalar equilibria

If $f_x(x^*; r) = 0$, the derivative test is inconclusive. The signs of $f(x; r)$ immediately to the left and right of x^* determine its stability.

Question: What is the state space?

Classify each equation as autonomous or nonautonomous and linear or nonlinear. Give an autonomous state space.

$$\dot{x} = x(1-x), \qquad \ddot{q} + q^3 = 0, \qquad \dot{y} = t - y.$$

Solution.

equation	autonomous	linearity	autonomous state
$\dot{x} = x(1-x)$	yes	nonlinear	$x \in \mathbb{R}$
$\ddot{q} + q^3 = 0$	yes	nonlinear	$(q, v) \in \mathbb{R}^2$
$\dot{y} = t - y$	no	linear nonhomogeneous	$(y, s) \in \mathbb{R}^2$

Question: What is the state space? (continued)

For the second and third equations,

$$\dot{q} = v, \qquad \dot{v} = -q^3,$$

and

$$\dot{y} = s - y, \qquad \dot{s} = 1, \qquad s(0) = 0,$$

respectively. The scalar equation $\dot{y} = t - y$ is linear nonhomogeneous, but its autonomous lift has the affine vector field

$$F(y, s) = (s - y, 1),$$

which is not linear because $F(0, 0) = (0, 1) \neq (0, 0)$.

Question: How does a repeated equilibrium affect a phase line?

Analyze

$$\dot{x} = (x + 1)^2(2 - x).$$

Solution. The equilibria are $x = -1$ and $x = 2$. Since $(x + 1)^2 > 0$ for $x \neq -1$,

x	$(-\infty, -1)$	-1	$(-1, 2)$	2	$(2, \infty)$
$f(x)$	$+$	0	$+$	0	$-$

and the phase line is

$$\longrightarrow\ \odot_{-1}\ \longrightarrow\ \bullet_2\ \longleftarrow\ .$$

Thus -1 attracts from the left and repels from the right; it is half-stable and Lyapunov unstable. The point 2 is asymptotically stable. Moreover,

$$\lim_{t\to\infty} x(t) = \begin{cases} -1, & x(0) \leq -1, \\ 2, & x(0) > -1. \end{cases}$$

Since $f'(2) = -9$, perturbations near 2 decay to leading order as e^{-9t}. Since $f'(-1) = 0$, linearization does not classify -1.

Question: How do phase line, potential, and basin agree?

For

$$\dot{x} = -x(x - 1)(x + 2),$$

find the equilibria, stability, basins, and a potential V satisfying $\dot{x} = -V'(x)$.

Solution. The sign table is

x	$(-\infty, -2)$	-2	$(-2, 0)$	0	$(0, 1)$	1	$(1, \infty)$
$f(x)$	$+$	0	$-$	0	$+$	0	$-$

Hence -2 and 1 are asymptotically stable, while 0 is unstable. Their basins are

$$\mathcal{B}(-2) = (-\infty, 0), \qquad \mathcal{B}(1) = (0, \infty).$$

Expanding $-f(x)$ gives

$$-f(x) = x(x - 1)(x + 2) = x^3 + x^2 - 2x,$$

Question: How do phase line, potential, and basin agree? (continued)

so one may take

$$V(x) = \frac{x^4}{4} + \frac{x^3}{3} - x^2.$$

Along solutions,

$$\dot{V} = V'(x)\dot{x} = -f(x)^2 \leq 0.$$

The asymptotically stable equilibria are local minima of V; the unstable equilibrium is a local maximum.

Visualization: Three local bifurcation signatures

With r as the parameter and x as the state,

type	normal form	equilibrium branches	structural condition
saddle-node	$\dot{x} = r - x^2$	$x = \pm\sqrt{r}\ (r \geq 0)$	no imposed symmetry or persistent branch
transcritical	$\dot{x} = rx - x^2$	$x = 0,\ r$	persistent branch $x = 0$
pitchfork	$\dot{x} = rx - x^3$	$x = 0,\ \pm\sqrt{r}\ (r \geq 0)$	$x \mapsto -x$ symmetry

Observation: Structural requirements for the three signatures

For unrestricted one-parameter scalar families, the generic equilibrium bifurcation is the saddle-node. Transcritical and pitchfork bifurcations are generic only within families that retain the indicated persistent branch or symmetry.

Question: Which local bifurcation occurs?

Find the equilibrium branches and their stability for

$$\dot{x} = \mu - (x-1)^2, \qquad \dot{x} = x(\mu - 2x), \qquad \dot{x} = \mu x - 2x^3.$$

Solution. For $f_1(x;\mu) = \mu - (x-1)^2$,

$$x_\pm = 1 \pm \sqrt{\mu} \quad (\mu \geq 0), \qquad (f_1)_x = -2(x-1).$$

For $\mu > 0$, x_- is unstable and x_+ is asymptotically stable. They collide at $(\mu, x) = (0, 1)$: a saddle-node.

For $f_2(x;\mu) = x(\mu - 2x)$,

$$x = 0, \qquad x = \frac{\mu}{2}, \qquad (f_2)_x = \mu - 4x.$$

The scalar linearized rates on the two branches are μ and $-\mu$, so the branches exchange stability at the origin: a transcritical bifurcation.

For $f_3(x;\mu) = \mu x - 2x^3$,

$$x = 0, \qquad x_\pm = \pm\sqrt{\frac{\mu}{2}} \quad (\mu > 0).$$

At the origin $(f_3)_x = \mu$, while

$$(f_3)_x(x_\pm) = \mu - 6\frac{\mu}{2} = -2\mu < 0.$$

Two asymptotically stable branches emerge as the origin loses stability: a supercritical pitchfork.

Question: What does an overdamped reduction forget?

Let $\varepsilon > 0$ and solve

$$\varepsilon y'' + y' = 1, \qquad y(0) = y_0, \qquad y'(0) = v_0.$$

Compare the solution with the reduced equation obtained by setting $\varepsilon = 0$.

Solution. Put $v = y'$. Then

$$\varepsilon v' + v = 1, \qquad v(0) = v_0,$$

so

$$v(t) = 1 + (v_0 - 1)e^{-t/\varepsilon}.$$

Integration gives

$$y(t) = y_0 + t + \varepsilon(v_0 - 1)(1 - e^{-t/\varepsilon}).$$

The reduced solution is $y_{\text{red}}(t) = y_0 + t$. On every bounded interval $0 \le t \le T$,

$$\sup_{0\le t\le T} |y(t) - y_{\text{red}}(t)| \le \varepsilon |v_0 - 1| \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Unless $v_0 = 1$, the velocities do not converge pointwise to $y'_{\text{red}}(t) = 1$ on $[0, T]$, because at $t = 0$,

$$v(0) - y'_{\text{red}}(0) = v_0 - 1.$$

They do converge at every fixed $t > 0$. Moreover, for every $0 < \delta \le T$,

$$\sup_{\delta\le t\le T} |v(t) - y'_{\text{red}}(t)| = |v_0 - 1|e^{-\delta/\varepsilon} \longrightarrow 0 \qquad (\varepsilon \downarrow 0),$$

so the convergence is uniform on $[\delta, T]$ but not on $[0, T]$ unless $v_0 = 1$. Since the exponential depends on t through t/ε, set $s = t/\varepsilon$. A fixed interval $0 \le s \le S$ corresponds to $0 \le t \le \varepsilon S$; this is the initial layer.

Question: How many decimal places does step refinement support?

An unknown quantity Q is approximated with step sizes $h, h/2, h/4$:

$$Q_h = 1.20384, \qquad Q_{h/2} = 1.20496, \qquad Q_{h/4} = 1.20524.$$

Assuming $p > 0$, $C \ne 0$, and

$$Q_h = Q + Ch^p + o(h^p) \qquad (h \downarrow 0),$$

estimate p, the finest-grid error, and Q.

Solution.

$$\frac{Q_{h/2} - Q_h}{Q_{h/4} - Q_{h/2}} = \frac{0.00112}{0.00028} = 4 = 2^2,$$

so $p \approx 2$. The finest-grid error is

$$|Q - Q_{h/4}| \approx \frac{|Q_{h/4} - Q_{h/2}|}{2^2 - 1} = \frac{0.00028}{3} \approx 9.33 \times 10^{-5}.$$

Richardson extrapolation gives

$$Q \approx Q_{h/4} + \frac{Q_{h/4} - Q_{h/2}}{3} = 1.205333\ldots.$$

Question: How many decimal places does step refinement support? (continued)

The data support three decimal places, provided the asymptotic error model is valid and roundoff is negligible.